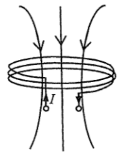
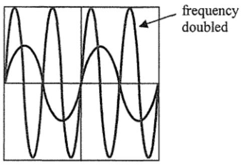
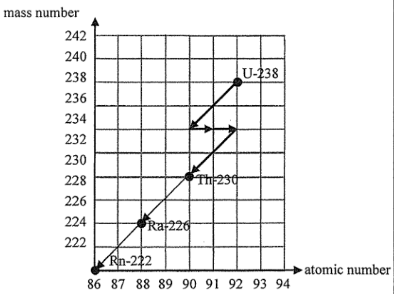
6.	(a)	(i)	Accelerates at g before the elastic cord stretches / at the beginning.	1A	
			Acceleration decreases as the cord stretches.	1A	
			Decelerates until momentarily at rest	1A	
			(after the tension in the cord is greater than mg).		3
		(ii)	Gravitational potential energy changed to kinetic energy and	1A	
			(then) elastic potential energy in elastic cord.	1A	2
	(b)		Elastic cord lengthens the stopping time,	1A	
			hence reduces the (net) force acting on the player.	1A	2
	(c)		Contact area is larger,	1A	
			hence pressure is smaller during the fall and the structure is less likely to break / detach.	1A	2

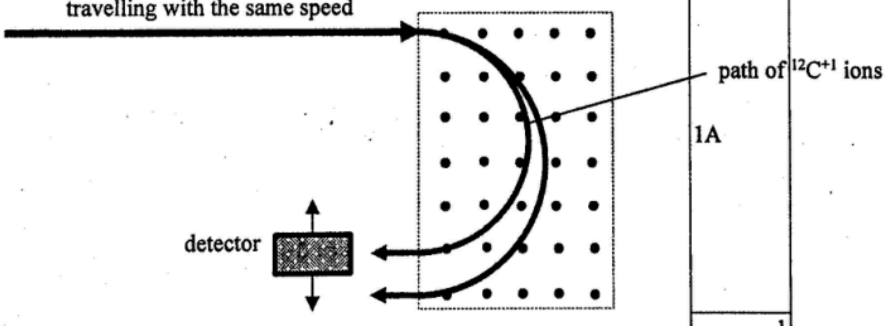
10. (a)	Alpha particles emitted can be stopped by the (thin) metallic casing.	1A	
	Or Shorter range / Lower penetrating power.	1A	
			1
(b) (i)	$k = \dfrac{\ln 2}{t_{1/2}} = \dfrac{\ln 2}{87.74 \times 3.16 \times 10^7}$	1M	
	$= 2.5 \times 10^{-10}\,\text{s}^{-1}$ or $7.9 \times 10^{-3}\,\text{year}^{-1}$		
	Activity $A = kN$		
	$= \dfrac{\ln 2}{87.74 \times 3.16 \times 10^7}(3.2 \times 10^{25})$	1M	
	$= 8.000 \times 10^{15}$ (Bq)	1A	
			3
(ii)	Power = Energy per decay × Activity		
	$= 5.5\text{ MeV} \times 8.000 \times 10^{15}\text{ Bq}$	1M	
	$= 5.5 \times 10^6 \times 1.60 \times 10^{-19} \times 8.000 \times 10^{15}$		
	$= 7040$ W or 7.040 (kW)	1A	
			2
(iii)	Power $\propto$ Activity		
	Activity $\propto N$		
	$\therefore$ Percentage of power left $= (\frac{1}{2})^{t/t_{1/2}} \times 100\%$		
	$= (\frac{1}{2})^{36/87.74} \times 100\%$	1M	
	$= 75.25\% \approx 75\%$	1A	
	Or Percentage / fraction of power left = 3/4		
	Alternatively: $N = N_0 e^{-kt}$		
	$\therefore$ Percentage of power left $= e^{-kt} \times 100\%$	1M	
	$= e^{-(\ln 2 \div 87.74) \times 36} \times 100\%$		
	$= e^{-0.2844} \times 100\%$		
	$= 75.25\% \approx 75\%$	1A	
			2

10. (a)	(i)	$226 - 206 = 20$ (multiple of 4 for α) $\therefore {}^{206}_{82}\text{Pb}$ is the end product	1M 1A	
			2	
	(ii)	% undecayed Ra-226 left $\dfrac{N}{N_0} = e^{-\frac{\ell n 2}{1600}\times 50}$ $= 97.857\ \%$ $\approx 97.9\ \%$	1M 1A	Accept 97.8 % to 98.0 %
		Or % undecayed Ra-226 left $= \left(\dfrac{1}{2}\right)^{\frac{50}{1600}}$ $\approx 97.9\ \%$	1M 1A	
			2	
(b)	(i)	$\because$ random process	1A	
			1	
	(ii)	Some of the daughter products of Ra-226 are also radioactive and may emit β particles.	1A	
			1	
	(iii)	Reason: weaker ionizing power of β and γ radiations	1A	
		- raise the source to a distance greater than the range of α (a few cm) will cease to have sparks. - insert a paper between the source and the gauze, sparks will cease to produce. Any ONE	1A	
			2	

9.	(a)	(i)	β decay / beta decay OR ${}^{40}_{19}\text{K} \rightarrow {}^{40}_{20}\text{Ca} + {}^{0}_{-1}\beta$	1A		
					1	
		(ii)	Justified: the penetrating power of β radiation enables it to penetrate the body's organ / skin Or Not justified: the activity is low and is comparable to background radiation / β radiation is largely shielded by the human body	1A		
					1	
	(b)	(i)	$\frac{0.45 \times 0.012\%}{40.0} = 1.35 \times 10^{-6}$ (mole)	1A		
					1	
		(ii)	$k = \frac{\ln 2}{1.25 \times 10^{9}\ (3.16 \times 10^{7})} = 1.754803 \times 10^{-17}\ (\text{s}^{-1})$	1M/1A		i. e. 5.545177×10^{-10} (year^{-1})
			Activity = kN $= 1.754803 \times 10^{-17} \times (1.35 \times 10^{-6} \times 6.02 \times 10^{23})$ $= 14.261284$ (Bq) ≈ 14.3 (Bq)	1A		Accept 14.2 to 14.3 (Bq)
					2	

12. (a) 	2A	1A for correct direction (also accept field lines pointing upwards for observers with different perception of current direction) 1A for correct field pattern
	2	
(b) (i) The a.c. flowing in transmitter coil T generates a changing magnetic field,	1A	
thus by electromagnetic induction an induced e.m.f. is produced in receiver coil R to oppose the changing magnetic flux experienced by it.	1A	
	2	
(ii) CRO display 	2A	1A for correct frequency 1A for correct amplitude
	2	
(c) If the back cover is metallic, eddy currents will be produced (by induction),	1A	
loss of energy / heating effect results / magnetic field / flux is blocked by the metal case and thus making wireless charging impossible.	1A	
<u>Or</u> Non-metallic materials such as glass are insulators, and no eddy currents are produced. As a result, energy loss / heating effect will be minimized / the magnetic field can pass through the back cover easily.		
	2	

Solution	Marks	Remarks
13. (a) (i) $k = \frac{\ln 2}{t_{½}}$		
$= \frac{\ln 2}{3.82 \times 24 \times 3600}$	1M	
$= 2.1001405 \times 10^{-6}\ \mathrm{s^{-1}} \approx 2.10 \times 10^{-6}\ (\mathrm{s^{-1}})$	1A	
	2	
(ii) $A = kN$		
$N = \frac{48}{2.10 \times 10^{-6}}$	1M	e.c.f. from (a)(i)
$= 2.285561 \times 10^{7} \approx 2.29 \times 10^{7}$	1A	Accept: $(2.28 \sim 2.3) \times 10^{7}$
	2	
(iii) As radon is a gas that can be inhaled into the lungs of human,	1A	
the relatively strong ionizing power of α particles emitted by decaying radon / radioactive gas might affect the organs / cells nearby.	1A	
	2	
(b) 	2A	
	2	

12. (a) (i) $^{14}_{6}C \rightarrow {}^{14}_{7}N + {}^{0}_{-1}e$ (or β)	1A	
	1	
(ii) $k = \dfrac{\ln 2}{t_{\frac{1}{2}}}$ $= \dfrac{\ln 2}{5370}$	1M	
$= 1.2097 \times 10^{-4}$ (yr^{-1}) $\approx 1.21 \times 10^{-4}$ (yr^{-1})	1A	
	2	
(iii) $C = C_0 e^{-kt}$ $6.0 = 13.6\, e^{-(1.21\times10^{-4})t}$	1M	e.c.f. from (a)(ii)
$t = 6764.7 \approx 6800$ (years)	1A	Accept: 6760 ~ 6820 (years)
		Or $(13.6)\left(\frac{1}{2}\right)^n = 6.0$ $n \log\left(\frac{1}{2}\right) = \log\left(\frac{6.0}{13.6}\right)$ $n = 1.181$ $t = n \times 5730 = 6764.7$ (years) ≈ 6800 (years)
	2	
(b) (i) An isotope is a variation of an element that possesses the same atomic number but a different mass number.	1A	Or same number of protons but different number of neutrons
	1	
(ii) 	1A	
	1	
(iii) As the magnetic force induced on a carbon ion is always perpendicular to its velocity / direction of motion,	1A	
no work is done on the ions.	1A	
	2	